

	rare	cortical blindness transient, neurotoxicities (e.g., somnolence and fatigue), optic neuritis, seizure***
Cardiac disorders	common	arrythmia, myocardial infarction
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Vascular disorders	common	hypertension, transient systolic hypotension following rapid intravenous administration
	uncommon	haemorrhage
Respiratory, thoracic and mediastinal disorders	rare	interstitial pneumonitis, pulmonary fibrosis
	not known	bronchospasm
Gastrointestinal disorders	very common	abdominal pain, anorexia, constipation, nausea and vomiting
	common	diarrhoea, mucositis (including stomatitis and esophagitis)
	rare	dysgeusia, dysphagia
Hepatobiliary disorders	very common	alanine aminotransferase increased, alkaline phosphatase increased, aspartate amino transferase increased, bilirubin increased, hepatotoxicity
Skin and subcutaneous tissue disorders	very common	alopecia, pigmentation
	common	pruritus, rash, urticaria
	rare	radiation recall dermatitis, Stevens- Johnsons syndrome, toxic epidermal necrolysis
Reproductive system and breast disorders	not known	infertility
General disorders and administration site conditions	very common	asthenia, malaise
	common	extravasation****, phlebitis
	rare	pyrexia
*Myelosuppression with fatal outcome has been reported **Anaphylactic reactions can be fatal ***Seizure is occasionally associated with allergic reactions. ****Post marketing complications reported for extravasation included local soft tissue toxicity, swelling, pain, cellulitis, and necrosis including skin necrosis.		

Hematological Toxicity: The dose-limiting effect of etoposide is myelosuppression. Bone marrow recovery is usually complete by day 20, and no cumulative toxicity has been reported. Granulocyte and platelet nadirs tend to occur about 10-14 days after administration of etoposide depending on the way of administration and treatment scheme. Nadirs tend to occur earlier with intravenous administration compared to oral administration. **Alopecia:** Reversible alopecia, sometimes progressing to total baldness has been observed in upto 66% of patients. **Blood pressure changes** **Hypotension:** Transient hypotension followingrapid intravenous

administration has been reported in patients treated with etoposide and has not been associated with cardiac toxicity or electrocardiographic changes. Hypotension usually responds to cessation of infusion of etoposide and/or other supportive therapy as appropriate. When resulting the infusion, a slower administration rate should be used.No delayed hypotension has been noted. **Hypertension:** In clinical studies involving etoposide injection, hypertension has been reported. If clinically significant hypertension occurs in patients receiving etoposide injection, appropriate supportive therapy should be initiated. **Allergic reactions:** Anaphylactic type reactions have also been reported to occur during or immediately after intravenous administration of etoposide. The role that concentration or rate of infusion plays in the development of anaphylactic type reactions is uncertain. Blood pressure usually normalizes within a few hours after cessation of the infusion. Anaphylactic type-reactions can occur with the initial dose of etoposide. Actual fatal reactions associated with bronchospasm have been reported with etoposide. Facial flushing was reported in 2% and skin rashes in 3% patients. **Metabolic complications:** Tumour lysis syndroms (sometime fatal) has been reported following the use of etoposide in association with other chemotherapeutic drugs. **Symptoms and Treatment of Overdose** Total doses of 2.4–3.5 g/m² of etoposide administered intravenously over 3 days have resulted in severe mucositis and myelotoxicity. Metabolic acidosis and severe hepatic toxicity have been reported in patients receiving higher than recommended intravenous doses of etoposide. There is no specific antidote available. Treatment should therefore be symptomatic and supportive, and patients should be closely monitored. Etoposide and its metabolites are not dialyzable. **Effects on Ability to Drive and Use Machine** No studies on the effects on the ability to drive and use machines have been performed. Etoposide may cause adverse reactions that affect the ability to drive or use machines such as fatigue, somnolence, nausea, vomiting, cortical blindness , hypersensitivity reactions with hypotension. Patients who experience such adverse reactions should be advised to avoid driving or using machines. **Preclinical Safety Data** Not Applicable **Instruction for Use** Etoposide Injection is intended for slow intravenous infusion. Etoposide should not be administered as a rapid intravenous injection. Route of administration: IV infusion Diluents: Sodium Chloride and Glucose Final Concentration: 0.2 mg/ml to 0.4 mg/ml Period of infusion: The infusion should be administered over a period of 30-60 minutes. **Method of administration:** Plastic devices made of acrylic or ABS (a polymer of acrylonitrile, butadiene and styrene) have been reported to crack or leak when used with undiluted Etoposide injection. Etoposide should only be given by slow intravenous infusion. Etoposide should not be administered by intrapleural or intraperitoneal injection. Etoposide must be diluted before administration. Resultant concentrations should not be greater than 0.4 mg/mL since precipitation can occur. Usually etoposide is added

to 250 mL of 0.9% sodium chloride or 5% glucose. The infusion should be administered over a period of 30–60 minutes. Chemical and physical in-use stability of the solution diluted to a concentration of 0.2 mg/ml and 0.4 mg/ml has been demonstrated in sodium chloride injection (0.9 % w/v) and glucose injection (5% w/v) for up to 96 hours at room temperature (30°C) respectively. From a microbiological point of view, the product should be used immediately. If not used immediately, in-use storage times and conditions prior to use are the responsibility of the user. Do not store the diluted product in a refrigerator (2-8°C) as this might cause precipitation. Use once only and any unused product and discard any residue with local requirements. **The use of a Pharmacy Bulk Pack should be restricted to suitably qualified pharmacists operating in suitably equipped hospital pharmacies or compounding centres. The Pharmacy Bulk Pack is intended for multiple dispensing into sterile solutions for subsequent infusion in individual patients in one treatment session. The Pharmacy Bulk Pack should be spiked only once.** **Incompatibilities:** Etoposide Injection should not be physically mixed with any other drug. Parenteral drugs should be inspected for particulate matter and discolouration prior to use. **Storage Condition** Store below 30°C. Keep the vial in the outer carton in order to protect from light. Discard unused portion. Do not refrigerate or freeze. **Shelf Life** **Unopened vial:** 36 months **After dilution:** Chemical and physical in-use stability of the solution diluted to a concentration of 0.2 mg/ml and 0.4 mg/ml has been demonstrated in sodium chloride injection (0.9 % w/v) and glucose injection (5% w/v) for up to 96 hours at room temperature (30°C) respectively. From a microbiological point of view, the product should be used immediately. If not used immediately, in-use storage times and conditions prior to use are the responsibility of the user. Do not store the diluted product in a refrigerator (2-8°C) as this might cause precipitation. **Therapeutic Code** ATC code: L01CB01 **Nature and contents of container:** Sedol (Etoposide Concentrate for Solution for Infusion) 20 mg/ml & 100 mg/5 ml is filled in 5 ml molded clear glass vial type-I with reinforced Non-PVC Base which are stoppered with 20 mm bromobutyl rubber stopper and tampered with 20 mm Aluminium flip off seal. Each vial is packed into secondary carton along with a package insert. **MAL No.: MAL22116021AZ** Product Registration Holder: **Unimed Sdn Bhd** 53, Jalan Tembaga SD 5/23, Bandar Sri Damansara, 52200, Kuala Lumpur, Malaysia Manufactured by: **VENUS REMEDIES LIMITED** Hill Top Industrial Estate, Jharmajri, EPIP Phase-I (Extn.), Bhatoli Kalan, Baddi, Distt. Solan, Himachal Pradesh-173205 (INDIA) **Date of Revision:** 30-11-2022



To be sold by retail on the prescription of a cancer specialist/ cancer hospital/ institution only

SEDOL

ETOPOSIDE CONCENTRATE FOR SOLUTION FOR INFUSION 20 mg/ml

Product Description
SEDOL - Etoposide Concentrate for Solution for Infusion 20 mg/ml
Description: Clear, colorless or slightly yellowish solution, free from visible extraneous particulate matter.
After dilution: Clear, colorless or slightly yellowish solution, free from visible extraneous particulate matter.
List of Compatible diluents: Sodium Chloride and Glucose
Each ml contains: Etoposide USP 20 mg.

Pharmacodynamics
Etoposide is a semi-synthetic podophyllotoxin derivative. Its main effect seems to occur during the G2 phase of the cell cycle. Two dose-dependent reactions occur: at high concentrations (>10 µg/ml), lysis can be observed of the cells entering mitosis; at low concentrations (0.3–10 µg/ml), the cells are prevented from entering the prophase. The main macromolecular effect appears to be inhibition of DNA synthesis.

Pharmacokinetics
Following intravenous infusion plasma levels of etoposide declines bi-exponentially with a distribution half-life of approximately 1.5 hours followed by a terminal elimination half-life of 4-11 hours. The total body clearance is 16–36 ml/minute/m² and is dose-independent within the dose interval 100-600 mg/m². The terminal half-life is also dose-independent in this dose interval. Etoposide does not accumulate in plasma when given as a daily intravenous administration of 100 mg/m² over 4–6 days. The steady state distribution volume is 7-17 L/m². The distribution to CSF is low and variable. In vitro studies shows that Etoposide is extensively bound to human plasma proteins (97%). The elimination of etoposide is both renal and non-renal. Following intravenous administration of 3H-etoposide (70-290 mg/m²) 42-67% of the dose was recovered in urine and 0-16% in faeces. 8-35% of the dose was excreted in the urine as unchanged drug within 24 hours. The main non-renal elimination route of etoposide is metabolism. Less than 6% of the given dose is excreted in the bile. In adults the total body clearance etoposide correlated with creatinine clearance, low serum albumin concentration and non-renal clearance. In children, elevated serum GPT has been associated with decreased etoposide clearance. Impaired hepatic or renal function may increase etoposide concentration in tissues due to metabolic and excretion routes.

Indication
- Small cell carcinoma of the lung.
- Acute monocytic and myelomonocytic leukaemia.
- Hodgkin’s disease.
- Non-Hodgkin’s lymphoma.

Recommended dose

Dosage: The usual dose of etoposide must be based on the clinical and haematological response and tolerance of the patient. A repeat course of etoposide should not be administered until the patient’s haematological function is within acceptable limits. **Adult:** The dosage for Etoposide Injection is 50-60 mg/m²/day intravenously for 5 consecutive days followed by a treatment free interval of 2-3 weeks. Total dose should not usually exceed 400 mg/m² per course. **Method of administration:** Plastic devices made of acrylic or ABS (a polymer of acrylonitrile, butadiene and styrene) have been reported to crack or leak when used with undiluted Etoposide injection. Etoposide should only be given by slow intravenous infusion. Etoposide should not be administered by intrapleural or intraperitoneal injection. Etoposide must be diluted before administration. Resultant concentrations should not be greater than 0.4 mg/mL since precipitation can occur. Usually etoposide is added to 250 mL of 0.9% sodium chloride or 5% glucose. The infusion should be administered over a period of 30-60 minutes. Chemical and physical in-use stability of the solution diluted to a concentration of 0.2 mg/ml and 0.4 mg/ml has been demonstrated in sodium chloride injection (0.9 % w/v) and glucose injection (5% w/v) for up to 96 hours at room temperature (30°C) respectively. From a microbiological point of view, the product should be used immediately. If not used immediately, in-use storage times and conditions prior to use are the responsibility of the user. Do not store the diluted product in a refrigerator (2-8°C) as this might cause precipitation. Use once only and any unused product and discard any residue with local requirements. **The use of a Pharmacy Bulk Pack should be restricted to suitably qualified pharmacists operating in suitably equipped hospital pharmacies or compounding centres. The Pharmacy Bulk Pack is intended for multiple dispensing into sterile solutions for subsequent infusion in individual patients in one treatment session. The Pharmacy Bulk Pack should be spiked only once.** **Dosage Adjustment** **Hepatic impairment:** Etoposide is contraindicated in severe hepatic dysfunction, and it should be used with caution in patients with mild to moderate hepatic impairment. **Renal impairment:** Since some etoposide (approximately 30%) is excreted unchanged in the urine, dosage adjustment may be needed in patients with impaired renal function. **Route of Administration** Intravenous **Contraindication** Etoposide is contraindicated in:

- Hypersensitivity to etoposide, podophyllotoxines or podophyllotoxin-derivatives or to any of the excipients
- Severe liver impairment
- Severe myelosuppression
- Breastfeeding
- Concomitant use of yellow fever vaccine or other live vaccines is contraindicated in immunosuppressed patients

Warning and Precautions Etoposide should be administered under the supervision of qualified physician experienced in the use of cancer chemotherapeutic agents. Physicians should be aware that the treatment with etoposide may be

an anaphylactic reaction manifested as chills, fever, flushing, tachycardia, bronchospasm, dyspnea and hypotension, which may be fatal. Treatment is symptomatic. The infusion must be stopped and followed by administration of pressor agents, corticosteroids, antihistamines or volume expanding resources by physician. There may be reactions at injection site during administration. If etoposide is to be administered intravenously, paravenous injection must be carefully avoided. It is recommended to monitor the infusion site closely for possible infiltration during drug administration. There is no known specific treatment for extravasation at this time. There may be severe myelosuppression with resultant infection or bleeding. There have been reports of fatal myelosuppression after administration of etoposide. Patients treated with etoposide should be monitored closely and frequently for myelosuppression both during and after treatment. Dose limiting bone marrow suppression is the most significant toxicity associated with etoposide treatment. The following observations should be made at the start of treatment and before each subsequent dose etoposide: platelet count, hemoglobin and total and differential count of leukocytes. If radiotherapy or chemotherapy was carried out before the start of etoposide treatment, a suitable interval must elapse for the bone marrow to recover. After the initial dose, subsequent doses adjusted if the neutrophil count below 500 cells/mm³ occurring in more than 5 days or associated with fever or infection, if there is platelet counts below 25,000 cells/mm³ if they develop any other toxicity of grade 3 or 4, or if the renal clearance is below 50 ml/min. The dosage should be modified to take into account the myelosuppressive effects of other drugs in the combination or the effects of previous radiation therapy or chemotherapy which may have compromised bone marrow reserve. Occurrence of acute leukemia, which can occur with or without a pre-leukaemic phase, has been reported rarely in patients treated with etoposide in combination with other antineoplastic drugs. Neither the cumulative risk, nor the predisposing factors related to the development of secondary leukaemia are known. The roles of both administration schedules and cumulative doses of etoposide have been suggested, but have not been clearly defined. An 11q23 chromosome abnormality has been observed in some cases of secondary leukaemia in patients who have received epipodophyllotoxins. This abnormality has also been seen in patients developing secondary leukaemia after being treated with chemotherapy regimens not containing epipodophyllotoxins and in leukaemia occurring de novo. Another characteristic that has been associated with secondary leukaemia in patients who have received epipodophyllotoxins appears to be a short latency period, with average median time to development of leukaemia being approximately 32 months. Before etoposide treatment is started, bacterial infections should be brought under control. The infusion should be given slowly, during 30 to 60 minutes, to avoid hypotension or bronchospasm. In all instances where the use of Etoposide is considered for chemotherapy, the physician must evaluate the need and usefulness of the drug against the risk of adverse reactions. Most such adverse reactions are reversible if detected early. If severe reactions occur, the drug should be reduced in dosage or discontinued and appropriate corrective measures should be taken according to the clinical judgement of the physician. Reinstitution of Etoposide therapy should be carried out with caution, and with adequate consideration of the further need for the drug and alertness as to possible recurrence of

toxicity. In patients with a lower serum albumin level, the risk of toxicity caused by etoposide can be elevated. Before the start of therapy, during the therapy, and before each course of treatment, a peripheral blood panel (white blood cells, platelets, haemoglobin), renal function, and hepatic function should be checked, and neurological functions should be investigated. Courses of therapy with etoposide should in generally be carried out only if the patient’s liver and kidneys are functioning normally. If the patient is suffering from hepatic or renal dysfunction, renal and hepatic function should regularly monitored due to the risk of accumulation. Furthermore, courses of therapy with etoposide should be carried out only if the peripheral nervous system is functioning normally. Etoposide is mutagenic and carcinogenic. This should be taken into account when a long-term treatment is performed. In view of etoposide mutagenic potential, both male and female patients use effective contraception during treatment and up to 6 months after treatment. It is recommended to seek genetic counseling if the patient wants to have children after treatment. Since etoposide may reduce fertility in men, may be considered to allow sperm storage for subsequent paternity. **Paediatric population:** Safety and efficacy in children has not been systematically studied. Anaphylactic reactions have been reported in paediatric patients who received Etoposide Injection Excipient (s) that the clinician should be aware of: **Ethanol:** Etoposide Injection contains 30.5% alcohol (ethanol), which corresponds to 240.64 mg of ethanol per ml of concentrate i.e up to 1.2 gm of ethanol per 5 ml vial, equivalent to 30 ml of beer or 12.55 ml of wine and up to 3 gm of ethanol per 12.5 ml vial, equivalent to 75 ml of bear or 31.4 ml of wine. There is a health risk to hepatic patients, alcoholics, epileptics, patients with organic brain diseases, pregnant women, breastfeeding women, and children, amongst others. The effect of other drugs may be reduced or increased. **Benzyl alcohol:** Because of the presence of benzyl alcohol, Etoposide Injection must not be given to premature babies or neonates. It may cause toxic and allergic reactions in infants and children upto 3 years old. **Polysorbate 80:** Etoposide Injection contains polysorbate 80. In newborn infants a life threatening syndrome of liver, cholestasis and renal failure, pulmonary deterioration, thrombocytopenia and ascites has been associated with an injectable vitamin E product containing polysorbate 80. **Interactions with Other Medicaments** The co-administration of high doses of cyclosporine (serum concentration >2000 ng/ml) and oral etoposide led to AUC values for etoposide that were elevated by 80% and to clearance that was reduced by 38% in comparison with etoposide monotherapy. Concomitant treatment with cisplatin is associated with reduced total body clearance of etoposide. Concomitant phenytoin or phenobarbital therapy is associated with increased etoposide clearance and reduced efficacy. Prior or concurrent use of other drugs with similar myelosuppression action as etoposide may be expected to have additive or synergetic effects. There is increased risk of fatal systemic vaccinal disease with the use of yellow fever vaccine. Live vaccines are contraindicated in

immunosuppressed patients. In vitro, plasma protein binding is 97%. Phenylbutazone, sodium salicylate, and acetylsalicylic acid may displace etoposide from plasma protein binding. Concomitant warfarin therapy may result in elevated international normalized ratio (INR). Close monitoring of INR is recommended. Cross resistance between anthracyclines and etoposide has been reported in preclinical experiments. **Pregnancy and Lactation** **Pregnancy:** Etoposide can cause fetal harm when administered to pregnant women. Etoposide have been shown to be teratogenic in mice and rats. There are no adequate and well controlled studies in pregnant women. Women of childbearing potential should be advised to avoid becoming pregnant. If the drug is used during pregnancy, or if the patient becomes pregnant while receiving the drug, the patient should be apprised of the potential hazard to the fetus. Given the mutagenic potential of etoposide, an effective contraception is required for both male and female patients during treatment and up to 6 months after ending treatment. Genetic consultation is recommended if the patient wishes to have children after ending the treatment. **Lactation:** It is not known whether these drugs are excreted in human milk. Because many drugs are excreted in human milk and because of the potential for serious adverse reactions in nursing infants from etoposide, a decision should be made whether to discontinue nursing or to discontinue the drug, taking into account the importance of the drug to the mother. Benzyl alcohol is probably excreted into breast milk and can be orally absorbed by the infant. **Fertility:** As etoposide may decrease male fertility, preservation of sperm may be considered for the purpose of later fatherhood. **Side Effects** No studies on the effects on the ability and use machines have been performed with etoposide. Fatigue, somnolence, nausea, vomiting and acute hypersensitivity reactions may occur due to a drop in blood pressure, and this may impair the ability to drive and use machines. The following adverse events have been reported in association with Etoposide therapy:

System Organ Class	Frequency	Adverse Reaction (MedDRA Terms)
Infections and infestations	common	infection
Neoplasms benign,malignant and unspecified (including cysts and polyps)	common	acute leukaemia
Blood and lymphatic system disorders	very common	anaemia, leucopenia, myelosuppression*, neutropenia, thrombocytopenia
Immune system disorders	common	anaphylactic reactions**
	not known	angioedema, bronchospasm
Metabolism and nutrition disorders	not known	tumour lysis syndrome
Nervous system disorders	common	dizziness
	uncommon	neuropathy peripheral